Description:

This hack combines the Robolamp remote hack with a "twist". Using only parts from other Hack Packs you can create *The Rotating Robolamp*. This hack creates base rotation for the robolamp. Base rotation allows your robot more freedom and flexibility to level up its assistance as your friendly light source. This hack also allows greater personality with moods seen best when in distracted mode; when distracted your robolamp will not just look away from what you need light on, it will rotate around to look at more interesting things than what you're working on. This hack includes Ir capabilities which allows the direction of light from a distance. Its sleek design allows for concealed battery packs while staying with the original Robolamp look. Including Ir abilities the lamp retains all button and joystick features seen in Robolamps stock code.

Required Materials/Sensors:

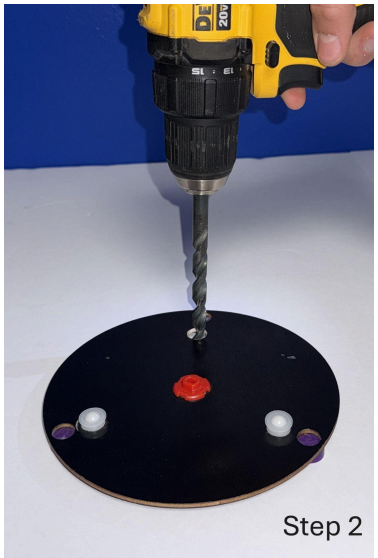
- RoboLamp
- Wooden base and platform piece from Card Dealer
- 15 short screw standoffs from Laser Tag
- 5 long screw standoffs from Label Maker
- Ir remote/sensor from Turret
- 1 gear motor from String Plotter
- Dual motor driver from String Plotter
- 3 male to male connectors from any robot
- 2 long metal M3 bolts from Card Dealer
- 2 metal M3 nuts from Card Dealer
- Battery pack from any Hack Pack.

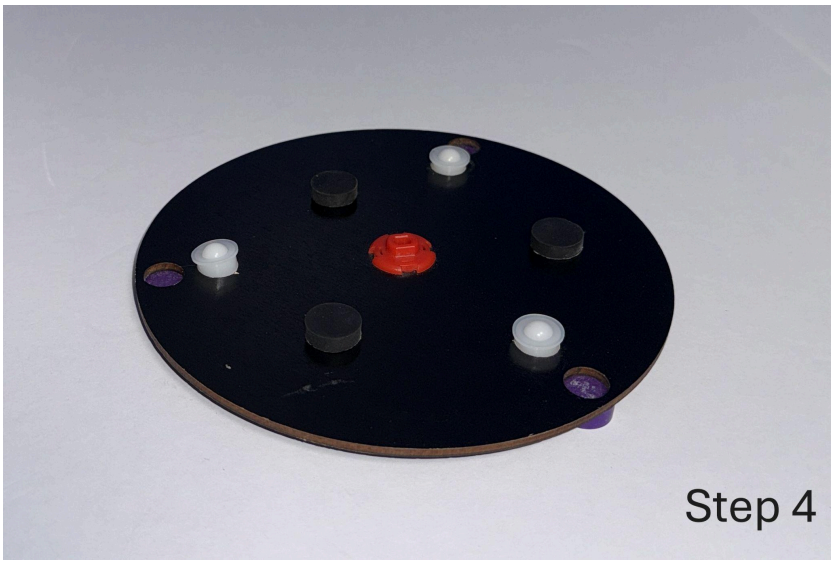
Required Tools:

- Drill
- Sandpaper

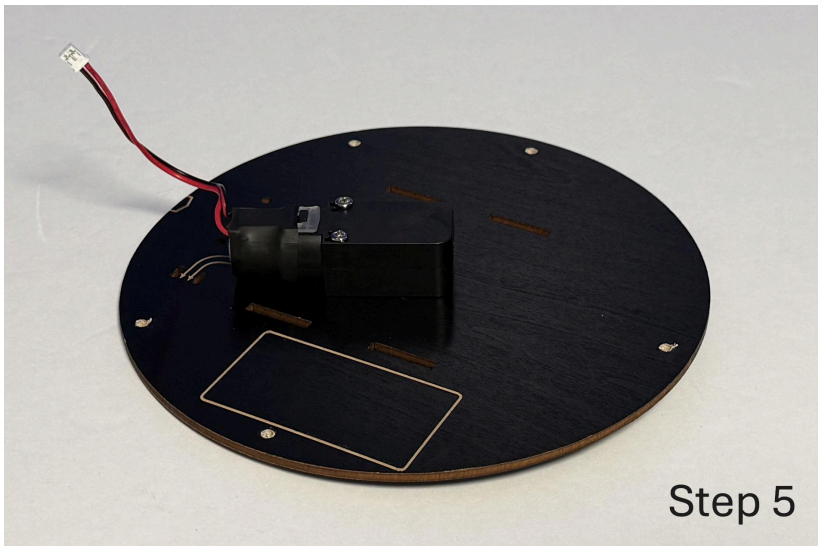
Assembly:

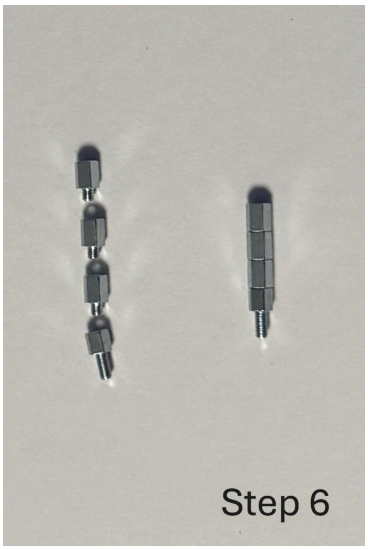
1. Grab the wooden base (Card Dealer), and pop out the transfer bearings.
2. In the wooden base, drill 7/16” holes ¼ inch away from the three transfer bearing hole (between the D-Shaft capture piece and original hole)
3. Widen the hole slightly by sanding the newly drilled holes until transfer bearings fit snuggly with sandpaper.
4. Place rubber feet from Robolamp in between the transfer bearings for stability.



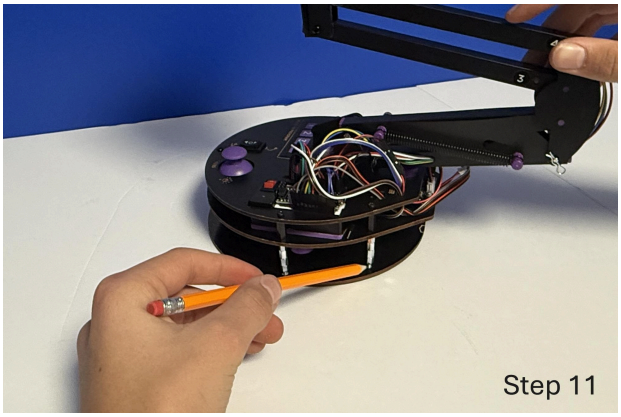


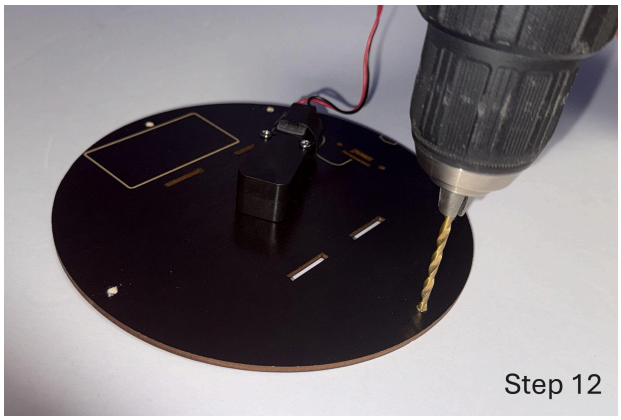
5. Get the wooden platform piece (Card Dealers) and gear motor (String plotter), align the gear motor to the wooden platform to thread 2 M3 nuts and bolts through holes provided.
6. Assemble together three short threaded standoffs (Laser Tag) and one long threaded standoff (Label Maker) making sure the long threaded standoff is visible on the bottom.



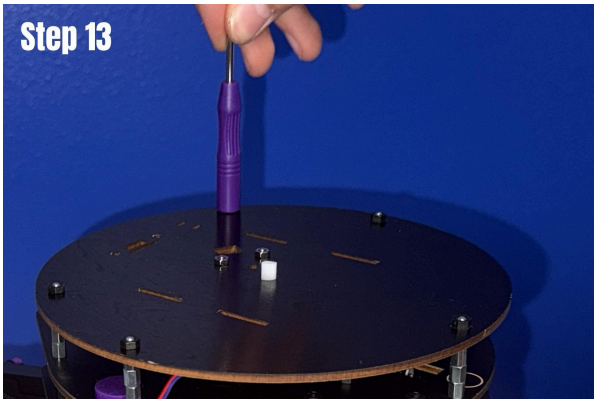


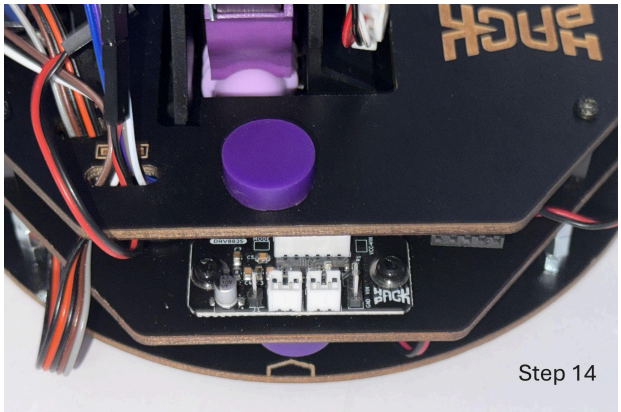
7. Repeat step 6, four more times for a total of five columns of standoffs.
8. Remove nuts from the base of the Robolamp.
9. Screw on the five standoff columns to the five open screws exposed in step 7.
10. Align the Robolamp with the platform piece so that the stepper motor's wires are towards the back of the Robolamp.
11. Mark where the standoff columns touch the platform piece.
12. Drill ⅛" holes in the platform piece you marked in step 11.



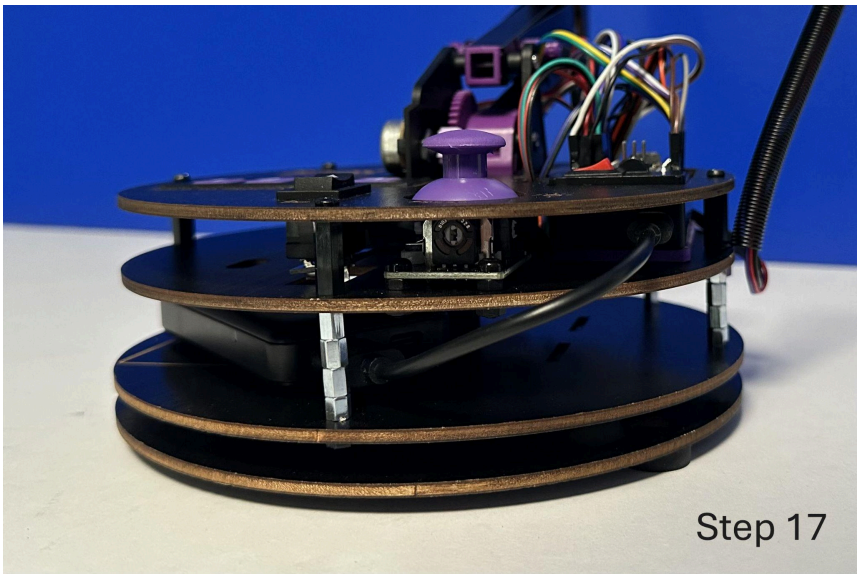


13. Connect platform piece to the Robolamp by sliding standoff columns through holes drilled and attach nuts to secure.
14. Replace the single motor driver with the dual motor driver (String Plotter).





15. Attach the Ir receiver antenna connected to jumper wires (Ir Turret) to the Robolamp wearever you like the look of.
16. Place Robolamp onto a wooden base.
17. Slide the battery pack under the joystick next to the rotational motor.



Wiring:

A0 Height Potentiometer (White)
A1 Joy button (Orange)
A2 Base motor DIR to motor driver (White)
A3 Lift motor PWM to motor driver (Grey)
A4 Base motor DIR to motor driver (Purple)
A5 Base motor PWM to motor driver (Blue)
A6 Joy Yaw joystick (Purple)
A7 Joy Pitch (Green)
D2 IR Receiver (Jumper wire color)
D3 Btn3 (Grey)
D4 Btn4 (Brown)
D5 Warm LED (Yellow)
D6 Cool LED (Green)
D7 Btn2 (Orange)
D8 Btn1 (Black)
D9 ServoPitch (Purple)
D10 ServoYaw (White)
D11 Height MotorPWM (Blue)
D12 Height MotorDIR (Purple)
D13 NeoPixel LED (Blue)
GND -> Common Ground to all devices (Black)
5V -> To all devices (Red)